

# GradeFrame | Bug Fix Log

BITS Digital CodeForge V1.0 - Verified 30 September 2026

| #   | Bug / Issue Identified                     | How You Reproduced It                                                                                                           | Root Cause                                                                    | Fix Implemented                                                                                                     | How You Tested the Fix                                                                                                 |
|-----|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| B01 | Reset fails before course selection        | Open the original app and confirm both Reset Range prompts before selecting a course. Browser harness reproduces the exception. | The handler dereferences grade controls before buildGradeUI has created them. | Disable Reset until a course has records; also guard the handler. Provide Undo instead of two confirmation prompts. | Browser regression: reset is disabled on initial load; reset after invalid edits restores valid boundaries and export. |
| B02 | Duplicate course options within one upload | Import the boundary fixture: four records belonging to Course A. The original creates four Course A options.                    | The upload handler adds an option for every record without deduplication.     | Build the course list from a Set of validated course names.                                                         | Browser regression: the same four-record XLSX produces exactly one course option.                                      |

Evidence: tests/core.test.cjs, tests/fixed-checks.html, tests/advanced-checks.html and tests/workspace-checks.html. Fixtures contain synthetic records.

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| #   | Bug / Issue Identified                             | How You Reproduced It                                                                                                            | Root Cause                                                                                 | Fix Implemented                                                                                                                                                                       | How You Tested the Fix                                                                                                                             |
|-----|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| B03 | Old courses survive replacement uploads            | Import the Course A fixture, then the one-record Course B fixture. Original options include four Course A entries plus Course B. | The upload handler appends options and replaces data without resetting dependent UI state. | A successful import atomically replaces data and dependent state only in the selected institute workspace. Failed imports retain its previous valid workbook with an explicit notice. | Browser regression: replacement leaves only Course B and updates its average to 65. Invalid replacement preserves the previous four valid records. |
| B04 | Minimum and maximum displayed under swapped labels | Import marks 0, 20, 80 and 100. Original browser output reads Min 100 and Max 0.                                                 | The Min label contains the max element ID; the Max label contains the min element ID.      | Render correctly named minimum and maximum fields from the shared statistics function.                                                                                                | Browser regression: Min 0, Max 100, mean 50 and median 50. Core tests independently check the expected values.                                     |

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|-----|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B05 | Incomplete grade coverage permits export | Set A maximum to 99, or E minimum to 1, in the original. Validation reports no error and export stays enabled even with boundary-mark students. | Validation checks adjacent bands but does not require the outer endpoints to be 100 and 0.                                | Validate complete 0-100 coverage in the core and at export time; lock the outer endpoint controls in the UI. | Core tests reject either missing endpoint, gaps and overlaps. Browser regression verifies locked endpoints and blocked invalid exports. All 101 integer marks are checked across defaults and 500 valid partitions. |
| B06 | Valid single-mark grade band is rejected | Set A to 100-100 and A-maximum to 99. Original validation rejects the otherwise continuous ranges.                                              | The condition $\text{min} \geq \text{max}$ disallows equality, although ranges are inclusive and marks are whole numbers. | Allow equal minimum and maximum while preserving continuity, ordering and endpoint checks.                   | Core tests assign 100 to A and 99 to A-. Browser regression accepts the single-mark band and updates its adjacent maximum.                                                                                          |

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|-----|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B07 | Empty course displays NaN statistics                       | After importing marks, choose the blank course option in the original. Mean and median become NaN, and export remains enabled with valid ranges. | Statistics divide by zero and index an empty array; export enablement depends only on range validity. | Return explicit null statistics for empty data and render dashes. Require records, valid ranges and a nonblank instructor for export. | Core empty-data test and initial browser empty-state check pass. Browser checks verify export blocking with no instructor and after clearing the instructor.                                             |
| B08 | File picker advertises only .xls despite .xlsx requirement | Inspect the original file input: accept is .xls. The challenge specifies .xlsx workbooks.                                                        | The HTML accept attribute names only the legacy extension.                                            | Accept .xlsx and .xls, validate the file signature and parse the workbook with a pinned local Excel reader.                           | Native file selection imports the four-record boundary XLSX. Browser regression rejects corrupt input. The actual downloaded CSV was reopened and all four expected IDs, marks and grades were verified. |

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